Table 1: *Test Problems, Dimensions and Initial Points.*

No.	Function	Dim.	Initial point
1	EPENLTY	100	$-6\mathbf{e}_{100}$
2	THUMP	2	$[-0.5; -0.5]$
3	THUMP	2	$[0.001; 0.001]$
4	EMARATOS	2	$21\mathbf{e}_2$
5	EMARATOS	30	$21\mathbf{e}_{30}$
6	DIAG5	10	$1.1\mathbf{e}_{10}$
7	DIAG5	50000	$1.1\mathbf{e}_{50000}$
8	DIAG5	100000	$1.1\mathbf{e}_{100000}$
9	TRECANNI	2	$[0.5; 0.5]$
10	TRECANNI	2	$[1.05; 1.05]$
11	QP1	4	$-\mathbf{e}_4$
12	QP1	100	$-\mathbf{e}_{100}$
13	ETRIDIAG1	50	$-0.5\mathbf{e}_{50}$
14	QP2	20	$-10\mathbf{e}_{20}$
15	QP2	5	$10\mathbf{e}_5$
16	QP2	10	$10\mathbf{e}_{10}$
17	QF1	5	$\mathbf{e}_5$
18	QF1	10	$\mathbf{e}_{10}$
19	QF2	50	$\mathbf{e}_{50}$
20	QF2	500	$\mathbf{e}_{500}$
21	QF2	1000	$\mathbf{e}_{1000}$
22	QF2	5000	$\mathbf{e}_{5000}$
23	POWER	2	$\mathbf{e}_2$
24	POWER	2	$15\mathbf{e}_2$
25	HIMMELBH	1000	$0.03\mathbf{e}_{1000}$
26	EHIEBERT	2	$[0.5; 0.5]$
27	SUMSQR	100	$0.1\mathbf{e}_{100}$
28	SUMSQR	2000	$0.1\mathbf{e}_{2000}$
29	SUMSQR	5000	$0.1\mathbf{e}_{5000}$
30	SHALLOW	1000	$10\mathbf{e}_{1000}$
31	SHALLOW	10000	$11\mathbf{e}_{10000}$
32	QUARTC	100	$0.23\mathbf{e}_{100}$
33	QUARTC	1000	$0.23\mathbf{e}_{1000}$
34	QUARTC	5000	$0.15\mathbf{e}_{5000}$
35	QUARTC	10000	$0.15\mathbf{e}_{10000}$
36	MATYAS	2	$[11; 11]$
37	MATYAS	2	$0.5\mathbf{e}_2$
38	DIAG1	5	$-10\mathbf{e}_5$
39	ETRIDIAG1	1000	$-5\mathbf{e}_{1000}$
40	DIAG1	5	$-5\mathbf{e}_5$
41	EHIEBERT	2	$[1.5; 1.5]$
42	ZAPF	2	$0.30\mathbf{e}_2$
43	EMARATOS	10	$21\mathbf{e}_{10}$
44	THUMP	2	$[0.1; 0.1]$
45	BD1	100	$0.5\mathbf{e}_{100}$
46	EMARATOS	50	$21\mathbf{e}_{50}$
47	RAYDAN2	10000	$0.3\mathbf{e}_{10000}$
48	RAYDAN2	50000	$0.3\mathbf{e}_{50000}$
49	RAYDAN2	100000	$0.3\mathbf{e}_{100000}$
50	FLETCHCR	10	$0.99\mathbf{e}_{10}$
51	FLETCHCR	1000	$0.99\mathbf{e}_{1000}$
52	FLETCHCR	50000	$0.99\mathbf{e}_{50000}$
53	DIAG3	5	$13\mathbf{e}_5$
54	DIAG3	10	$13\mathbf{e}_{10}$
55	DIAG3	2	$11\mathbf{e}_2$
56	EDENSCHNB	100	$0.03\mathbf{e}_{100}$
57	EDENSCHNB	5000	$-0.03\mathbf{e}_{5000}$
58	EDENSCHNB	10000	$-0.03\mathbf{e}_{10000}$
59	DIAG6	10000	$-1.01\mathbf{e}_{10000}$
60	DIAG6	5000	$-1.01\mathbf{e}_{5000}$
61	DIAG6	10000	$-1.03\mathbf{e}_{10000}$
62	DIAG6	50000	$-1.03\mathbf{e}_{50000}$
63	DIAG4	1000	$\mathbf{e}_{1000}$
64	DIAG4	10000	$\mathbf{e}_{10000}$
65	DIAG4	100000	$\mathbf{e}_{100000}$
66	DIAG7	10	$\mathbf{e}_{10}$
67	DIAG7	50	$\mathbf{e}_{50}$
68	DIAG7	100	$\mathbf{e}_{100}$
69	DIAG8	100	$-0.5\mathbf{e}_{100}$
70	DIAG8	500	$-0.5\mathbf{e}_{500}$
71	QUARTICM	1000	$0.03\mathbf{e}_{1000}$
72	QUARTICM	10000	$0.03\mathbf{e}_{10000}$
73	LPERTURBED	5000	$13\mathbf{e}_{5000}$
74	LPERTURBED	10000	$13\mathbf{e}_{10000}$
75	LPERTURBED	20000	$13\mathbf{e}_{20000}$
76	TWH	2	$[-15; -15]$
77	TWH	2	$[-25; -0.25]$
78	ENGVAL1	2	$-1.01\mathbf{e}_2$
79	ENGVAL1	2	$-1.03\mathbf{e}_2$
80	ENGVAL8	4	$0.003\mathbf{e}_4$
81	ENGVAL8	2	$0.003\mathbf{e}_2$
82	DENSCHNF	1000	$0.001\mathbf{e}_{1000}$
83	DENSCHNF	10000	$0.001\mathbf{e}_{10000}$
84	DENSCHNF	50000	$0.001\mathbf{e}_{50000}$
85	SIXHUMP	2	$[19; 19]$
86	SIXHUMP	2	$[0.01; 0.01]$
87	PRICE4	2	$-0.03\mathbf{e}_2$
88	PRICE4	2	$-0.01\mathbf{e}_2$
89	ZAPF	2	$0.31\mathbf{e}_2$
90	ZAPF	2	$0.29\mathbf{e}_2$
91	BD1	1000	$0.5\mathbf{e}_{1000}$
92	EXHIMMELBLAU	1000	$-5\mathbf{e}_{1000}$
93	EXHIMMELBLAU	2000	$-5\mathbf{e}_{2000}$
94	REF	2	$[-0.0150; -0.0001]$
95	REF	2	$[0.03; 0.009]$
96	EVF	2	$[11; 11]$
97	EVF	2	$13\mathbf{e}_2$
98	QP1	200	$-\mathbf{e}_{200}$
99	EPENLTY	1000	$-6\mathbf{e}_{1000}$
100	EPENLTY	20000	$-6\mathbf{e}_{20000}$

Note: $\mathbf{e}$ denotes the n-dimensional vector of ones, $\mathbf{e} = (1, \ldots, 1)^\top \in \mathbb{R}^n$.

Table 2: *Numerical results of DSPRP, GSPRP, DPRP, and MZDK on 100 test problems.*

	DSPRP			GSPRP			DPRP			MZDK		
No.	NI	NFE	CPU TIME	NI	NFE	CPU TIME	NI	NFE	CPU TIME	NI	NFE	CPU TIME
F1	5	19	0.0003399	6	20	0.0000928	6	24	0.000078	6	20	0.0001172
F2	8	75	0.000076	14	299	0.0001451	20	321	0.0001122	12	236	0.0001557
F3	5	33	0.0001283	8	167	0.0001319	17	186	0.0000547	7	145	0.0001303
F4	12	67	0.0001061	26	124	0.0001272	38	185	0.0000933	24	120	0.0002727
F5	16	74	0.0001011	28	128	0.0000869	81	369	0.0001029	22	107	0.0000867
F6	2	8	0.0016367	2	8	0.0081947	5	12	0.0030578	2	8	0.0028556
F7	2	11	0.0795431	2	8	0.0626196	5	12	0.3860886	2	8	0.1563884
F8	2	11	0.1976702	2	8	0.2657991	5	12	0.6795074	2	8	0.195499
F9	6	14	0.0001768	11	24	0.0000532	6	15	0.0000688	6	14	0.0001388
F10	7	17	0.0002142	8	19	0.0000584	4	23	0.0001137	7	17	0.0001571
F11	6	13	0.0023088	11	24	0.0092668	8	21	0.0070053	7	16	0.0029928
F12	9	42	0.0038188	11	23	0.0033256	8	24	0.0054037	7	15	0.0051393
F13	9	44	0.000211	15	60	0.0001023	37	232	0.0002983	8	47	0.0002217
F14	3	17	0.0025227	3	16	0.0059252	7	52	0.0156794	3	16	0.0052912
F15	4	19	0.0039752	4	18	0.0041739	4	20	0.0084792	4	18	0.0053167
F16	4	19	0.0145685	1000	50784	3.0292367	4	20	0.0084512	1000	50784	3.0003766
F17	10	78	0.0254262	13	202	0.0247342	2	7	0.0068144	13	202	0.0118639
F18	11	87	0.0114302	21	373	0.0198734	2	9	0.0053099	21	373	0.0349028
F19	2	5	0.0031641	2	5	0.0036424	5	11	0.0031846	2	5	0.0046626
F20	1	3	0.0022882	1	3	0.0016695	2	6	0.0058832	1	3	0.0037254
21	1	3	0.0017062	1	3	0.0038038	2	6	0.0034111	1	3	0.0026218
F22	1	3	0.0027755	1	3	0.0048004	2	6	0.0079951	1	3	0.0029042
F23	6	18	0.0000743	6	18	0.0000589	2	5	0.0000913	6	18	0.000075
F24	6	18	0.0001109	6	18	0.0000572	2	5	0.0000559	6	18	0.000075
F25	11	25	0.000409	12	40	0.0004504	6	13	0.0004104	9	18	0.0002375

Table 2: *Numerical results of DSPRP, GSPRP, DPRP, and MZDK on 100 test problems (continued).*

	DSPRP			GSPRP			DPRP			MZDK		
No.	NI	NFE	CPU TIME	NI	NFE	CPU TIME	NI	NFE	CPU TIME	NI	NFE	CPU TIME
F26	22	99	0.0002444	59	235	0.000081	1000	3509	0.0000631	26	99	0.0002128
F27	7	87	0.0001116	9	144	0.0000997	20	91	0.0001257	9	144	0.0000805
F28	9	227	0.0004816	12	276	0.0001922	10	150	0.0001585	12	276	0.0001864
F29	11	416	0.0005331	13	377	0.0003869	11	230	0.0004921	13	377	0.0004968
F30	25	95	0.0004076	16	51	0.0002408	21	109	0.0003269	12	45	0.00026
F31	17	56	0.0012556	23	78	0.0011836	15	90	0.0010683	29	79	0.0025721
F32	2	10	0.0048984	4	24	0.0032225	4	19	0.0046808	4	24	0.0071375
F33	2	10	0.0087547	4	24	0.0244704	6	33	0.0809098	4	24	0.0247468
F34	6	32	0.032922	3	17	0.0459508	6	34	0.0778978	3	17	0.0484285
F35	6	31	0.0619697	3	17	0.0351927	6	34	0.1852544	3	17	0.0744266
F36	1	7	0.0003169	1	7	0.0001605	2	9	0.0001299	1	7	0.0004485
F37	1	7	0.0002018	1	7	0.0002019	2	9	0.0002167	1	7	0.0002797
F38	22	57	0.0054262	21	57	0.0048322	10	43	0.012501	22	48	0.0048241
F39	11	49	0.0004667	8	31	0.0009373	16	94	0.0009023	17	67	0.0009008
F40	19	45	0.0060599	18	45	0.0128997	11	30	0.0068077	17	37	0.0086015
F41	35	139	0.0001183	42	174	0.0001434	1000	3387	0.0000593	27	105	0.0002408
F42	6	17	0.0001269	6	13	0.0001086	7	17	0.0001211	4	8	0.0002185
F43	13	74	0.0001016	28	128	0.0001691	82	378	0.0001503	24	114	0.0000948
F44	9	122	0.0001014	13	280	0.0000584	25	353	0.0000448	8	99	0.000083
F45	12	28	0.0001683	13	42	0.0002286	13	62	0.0001988	12	37	0.0002702
F46	14	66	0.0001478	28	128	0.0001897	80	365	0.0001111	1000	50083	0.0000836
F47	2	4	0.0068647	3	4	0.0059478	4	10	0.0400415	3	4	0.0095899
F48	2	4	0.0270687	3	4	0.0240568	4	10	0.1769232	3	4	0.0248041
F49	2	4	0.0497356	3	4	0.076627	4	10	0.3882985	3	4	0.1038959
F50	2	6	0.0001095	2	6	0.0004949	3	12	0.0001587	2	6	0.0002534

Table 2: *Numerical results of DSPRP, GSPRP, DPRP, and MZDK on 100 test problems (continued).*

	DSPRP			GSPRP			DPRP			MZDK		
No.	NI	NFE	CPU TIME	NI	NFE	CPU TIME	NI	NFE	CPU TIME	NI	NFE	CPU TIME
F51	2	6	0.0001713	2	6	0.0008419	3	12	0.0001985	2	6	0.0003704
F52	2	6	0.0046953	3	8	0.0153669	3	12	0.0109996	3	8	0.0115353
F53	16	46	0.0063349	20	46	0.0089483	82	398	0.06534	16	36	0.0094433
F54	19	53	0.006983	24	58	0.0156783	145	591	0.2021426	22	50	0.0164839
F55	12	32	0.0033177	9	21	0.0131952	6	24	0.0094588	7	15	0.0041632
F56	6	10	0.0000928	10	20	0.0002155	8	19	0.0001936	5	10	0.0005492
F57	5	15	0.0007147	5	10	0.0012284	6	14	0.0010419	5	10	0.0018374
F58	5	15	0.0012482	6	12	0.0050487	6	14	0.0026582	5	10	0.0021367
F59	2	5	0.0058301	3	5	0.0131955	5	13	0.0946156	3	5	0.0235591
F60	2	5	0.0044722	3	5	0.0159604	5	13	0.0369421	3	5	0.0191164
F61	2	5	0.0092428	3	5	0.024963	5	13	0.0982303	3	5	0.0197991
F62	2	5	0.0305305	3	5	0.0605619	5	13	0.1974715	3	5	0.0735764
F63	18	41	0.0068316	8	16	0.0071043	2	5	0.0048155	7	12	0.004893
F64	15	35	0.0183942	8	16	0.0237293	2	5	0.0177042	9	16	0.0126028
F65	5	17	0.1155724	8	16	0.2829493	2	5	0.1432154	10	17	0.2845935
F66	4	7	0.0016005	3	6	0.0030441	5	12	0.0075794	3	6	0.0049453
F67	3	7	0.0015592	3	6	0.0088325	5	12	0.014376	3	6	0.0043224
F68	3	7	0.0023551	3	6	0.0024635	5	12	0.0097244	3	6	0.0056145
F69	4	8	0.001808	3	7	0.005193	3	10	0.0094527	3	7	0.0047773
F70	3	8	0.0021609	3	7	0.0057474	3	10	0.0101818	3	7	0.0051533
F71	4	13	0.0006002	4	31	0.001025	4	25	0.0007706	4	31	0.0011647
F72	4	13	0.0022623	5	48	0.0030177	4	25	0.0049107	5	48	0.0026163
F73	2	4	0.0006422	2	4	0.0006488	2	6	0.0008219	2	4	0.0012486
F74	3	6	0.0011449	2	4	0.0019091	2	6	0.0021692	2	4	0.000768
F75	3	6	0.0017597	3	6	0.003112	2	6	0.004227	3	6	0.0023098

Table 2: *Numerical results of DSPRP, GSPRP, DPRP, and MZDK on 100 test problems (continued).*

	DSPRP			GSPRP			DPRP			MZDK		
No.	NI	NFE	CPU TIME	NI	NFE	CPU TIME	NI	NFE	CPU TIME	NI	NFE	CPU TIME
F76	23	105	0.0001234	37	188	0.0002985	227	861	0.000105	37	203	0.0001889
F77	11	58	0.0001113	112	278	0.0002705	24	160	0.0001157	12	66	0.0000888
F78	8	20	0.0003284	10	26	0.0001892	9	31	0.000091	8	20	0.0001645
F79	7	17	0.0002013	10	26	0.0002803	9	31	0.0000993	9	23	0.0001377
F80	11	23	0.000122	31	411	0.0001534	27	127	0.0001406	27	215	0.0001073
F81	9	54	0.0001251	29	440	0.0003595	5	20	0.0002405	30	381	0.0001212
F82	10	49	0.0002697	16	58	0.0012244	13	41	0.0004672	11	46	0.0003039
F83	13	55	0.0017492	17	58	0.0029591	13	41	0.0046785	12	48	0.0020222
F84	10	56	0.0079117	16	56	0.0140182	13	41	0.0139026	12	48	0.0116715
F85	7	42	0.0002095	9	49	0.0000999	18	83	0.0001053	9	51	0.0000703
F86	9	26	0.0001067	9	30	0.0001214	13	50	0.0000732	9	23	0.00024
F87	2	5	0.0000936	2	5	0.0004611	3	9	0.0001899	2	5	0.0001381
F88	2	4	0.0001054	2	4	0.0002817	2	5	0.0001359	2	4	0.0002221
F89	6	18	0.0001072	6	13	0.0001663	6	15	0.0000924	4	8	0.0001815
F90	6	18	0.0001055	6	14	0.000393	7	18	0.0001074	5	10	0.0000743
F91	13	31	0.0002515	15	48	0.0006889	14	65	0.0007023	15	48	0.0005507
F92	21	458	0.0770387	45	1026	0.3147663	21	264	0.203615	46	1030	0.1648155
F93	22	482	0.126911	46	1047	0.6448631	20	272	0.3108363	42	1003	0.2280491
F94	40	809	0.0001593	33	737	0.0002892	57	1882	0.0000728	27	646	0.0001031
F95	41	856	0.000153	50	1377	0.0002183	41	1227	0.0000746	30	672	0.0000878
F96	14	92	0.0001922	12	71	0.0001722	14	81	0.0000732	10	57	0.0000716
F97	10	49	0.0000811	16	76	0.000244	14	68	0.0000715	12	69	0.000183
F98	9	23	0.002802	12	25	0.014301	6	16	0.01008	7	15	0.0058729
F99	6	20	0.0001734	6	19	0.0005699	6	23	0.000205	6	19	0.000228
F100	5	16	0.0017268	1000	50762	0.0014764	4	20	0.0028355	1000	50762	0.003324